A. Installation Guide

1. Extract the ZIP file

- Locate the **ZIP file** you downloaded for this project.
- Right-click on the ZIP file and select "**Extract All...**" (Windows) or "**Extract Here**" (Mac/Linux).
- Choose an **easy-to-find folder** (such as your Desktop or Documents folder) as the destination.
- After extraction, **open the extracted folder** to ensure all files and subfolders are present.

In Jupyter Notebook, run Step 2 to Step 4:

2. Data Pre-processing

- **Open Jupyter Notebook**
- In the Jupyter file browser, navigate to the extracted project folder.
- Open the notebook named **Data Pre-processing.ipynb** in Source Codes folder.
- **Run all cells** from the beginning to the end.
- The notebook will process the data and generate a cleaned CSV file in Data folder:
  - **Dengue_QA_multilingual_cleaned.csv**
- Confirm that this file appears in your folder when processing is complete.

3. Knowledge Graph

- Open the notebook named **Knowledge Graph.ipynb** in Source Codes folder.
- **Run all cells** in this notebook.
- Upon completion, two files will be created in your Data folder:
  - **Dengue_QA_multilingual.json**
  - **Dengue_QA_multilingual_KG.ttl**
- Ensure these files appear in the project directory.

4. Model Training

- In Jupyter, open and run each of the following notebooks in Source Codes folder:
  - **Model Training (mpnet-base-v2).ipynb**
  - **Model Training (MiniLM-L12-v2).ipynb**

- Run all cells in each notebook. These will train the respective models.
- When finished, ensure the trained model and evaluation results files are created and saved in your Data folder:
    - **dengue_qa_model (MiniLM-L12-v2)**
    - **evaluation_results (MiniLM-L12-v2)**
    - **dengue_qa_model (mpnet-base-v2)**
    - **evaluation_results (mpnet-base-v2)**

5. **Deploying the Model on Hugging Face Spaces**
    - **Go to Hugging Face Spaces**
        - Open your browser and go to: https://huggingface.co/spaces
    - **Create a New Space**
        1. Click "**New Space**" at the top right.
        2. Enter **Space Name**: Lab1B-G5
        3. Enter **Short Description**: Briefly describe your project.
        4. Select Space SDK: Choose **Gradio**.
        5. Choose Gradio Template: Select **blank**.
        6. Space Hardware: Keep the default option (**free version** is sufficient).
        7. Select Visibility: Set to **Public** so anyone can access your app.
        8. Click "**Create Space**".
    - **Upload Project Files to Your Space**
        1. Once your space is created, click on the Files tab at the top-right.
        2. Upload the following files:
            - **app.py** (in Source Codes folder)
            - **requirements.txt** (in Source Codes folder)
            - **dengue_qa_model (mpnet-base-v2).pkl** (in Data folder)
    - **Run the Application**
        1. After uploading, the system will **automatically build and start your app**.
        2. Wait for the status to change to "**Running**" (this may take a few minutes).
        3. When running, click the "**App**" tab at the top. You will see your deployed web interface and can interact with your system.

Note: *If you encounter any errors, check the* ***Logs*** *tab in Hugging Face Spaces for troubleshooting details.*

B. How to Use the QA System

1. Interface

- **Home Page:**

  When you access the deployed app (via the App tab on Hugging Face Spaces), you'll see a web page titled "Sistem Soal Jawab Denggi" at the top.

- **Chatbox:**

  The main feature on the page is a chatbox, where you can type your questions.

2. Asking Questions

- **Multilingual Support:**

  You can ask questions in **Malay**, **English**, or a mix of both ("**rojak**" style). The system is designed to understand and respond to all three.

- **System Support Scope:**

  - ✓ **Apakah itu denggi** (What is dengue)
  - ✓ **Gejala/Simptom** (Symptoms of dengue)
  - ✓ **Rawatan dan penjagaan** (Treatment and care for dengue)
  - ✓ **Cara pencegahan** (Ways to prevent dengue)

  You are encouraged to ask any questions related to these topics for the most accurate and relevant answers.

- **How to Ask:**

  1. Click on the **input box** at the bottom of the chat interface.
  2. **Type your question** in your preferred language.
     - *Malay* example*:*
       Apakah simptom denggi?
     - *English* example*:*
       What are the symptoms of dengue?
     - *Rojak* example*:*
       Denggi symptom macam mana?
  3. Click the **"Hantar" button** (or press Enter) to submit your question.

3. Viewing Answers

- **Automated Response:**

  The system will process your question and display an answer just below your message in the chatbox.

- **Language Consistency:**

The reply will be in the **same language** (or style) as your question, whether it's Malay, English, or mixed.

- **Reliable Information:**
  All answers are generated based on the **knowledge graph**, ensuring that the information provided is accurate, reliable, and relevant to your question.

Quick Tips:

- **Keep your questions short and clear** for best results.
- **If the system doesn't understand your question**, try rephrasing or using simpler language.